獨孤印

Dugu Seal

The 26-faced jet seal of Dugu Xin of Western Wei

INTERACTIVE 3D RECONSTRUCTION

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JavaScript + Three.js + WebGL 2

Single HTML File · Runs on Native Browser · Easy Compilation

01

Browser as Engine

Three.js encapsulates modern WebGL 2 features, including programmable shaders, floating-point framebuffers, and screen-space derivatives (dFdx/dFdy). All rendering runs natively in the browser. In this project, Three.js serves as the 3D rendering engine, handling the construction of the 26-faced geometry and real-time lighting and shading.

02

Single-File Deployment

Geometry, materials, shaders, and UI interaction logic are all integrated into a single HTML document, with dependencies (Three.js) loaded via CDN for easy deployment and archival. A class-based architecture is used to manage seal logic, including the stamping charge system and multi-angle camera control.

03

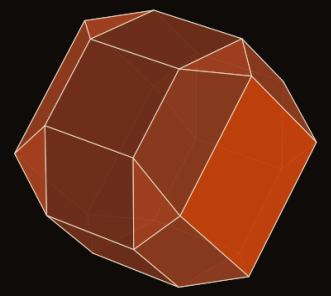
Highly Customizable

Using *onBeforeCompile*, the standard PBR shader is extended without breaking the Three.js lighting pipeline. GLSL enables procedural generation of artifact materials, engravings, and wear effects, while the Canvas 2D API simulates ink blending on paper at the pixel level.

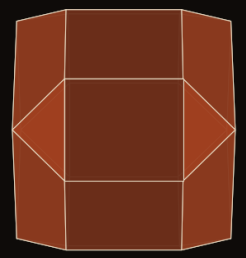
Truncated Skew Half-Cube

Rhombicuboctahedron · Archimedean Solid

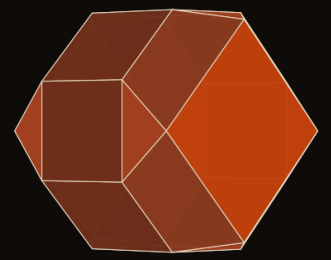
Isometric



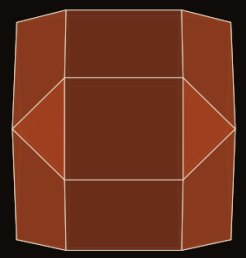
Front (corner)

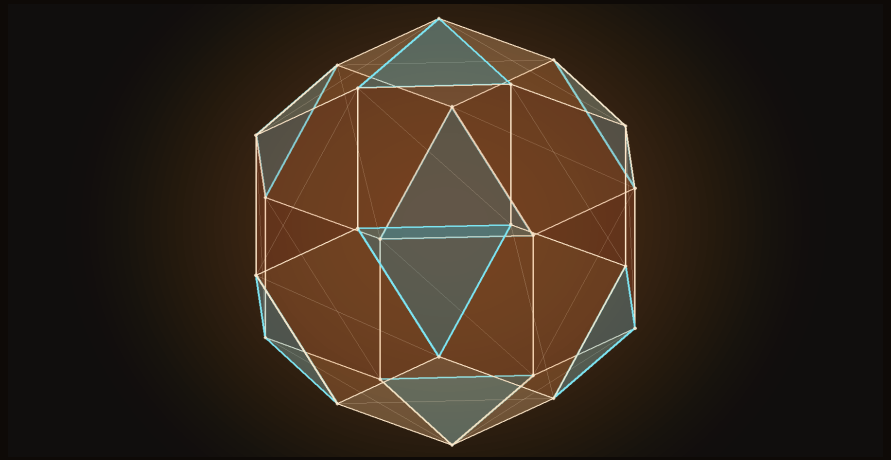


Front (edge)



Top





24	48	26
VERTICES 点	EDGES 棱	FACES 面

$$\chi = V - E + F = 24 - 48 + 26 = \mathbf{2} \quad \checkmark$$

Construction Method

The vertices are generated from all permutations of **$(\pm 1, \pm 1, \pm (1+\sqrt{2}))$**

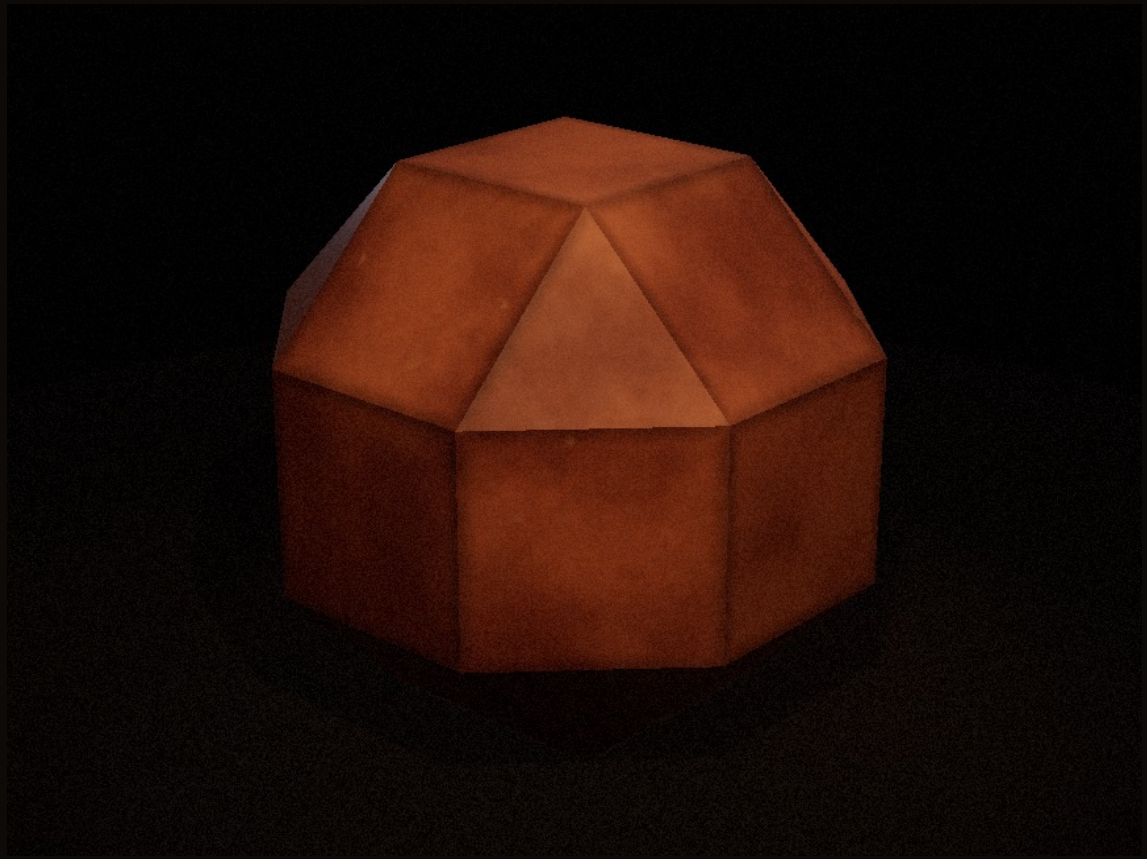
The 26 faces are classified into four types: 2 square caps, 8 belt squares, 8 longitudinal rectangles, and 8 equilateral triangles, forming 18 squares and 8 triangles in total.

A 6×5 dynamic texture matrix is designed to unwrap the 26 physical faces and map them one-to-one onto a 30-cell UV grid.

Each face is assigned an independent vertex set to ensure decoupling between geometric normals and UV coordinates.

Physically Based Rendering of Lacquerware Texture

Lighting Effects · Material Design · Age Rendering



Unengraved Rendering - Material Showcase

Museum-Quality Lighting

Physically Based Light Simulation

A three-point lighting model (warm key, cool fill, rim light) combined with GLSL-based HDR environment generation achieves museum-grade dynamic lighting and realistic reflections.

Customized Material Design

Custom GLSL Material Shader

Using onBeforeCompile, the standard PBR shader is extended with FBM noise and layered procedural textures, enabling real-time simulation of jet stone micro-textures and transitions between red lacquer and wood substrate.

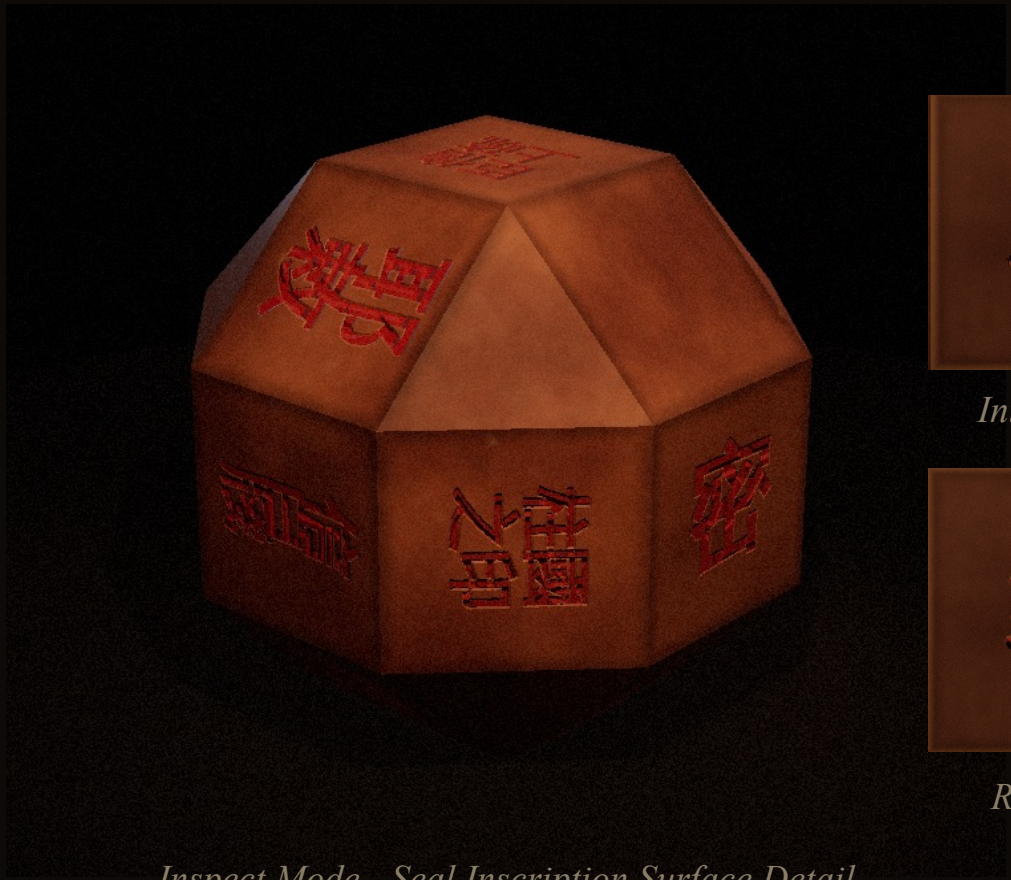
Quantifying Time's Texture

Age in Mathematics

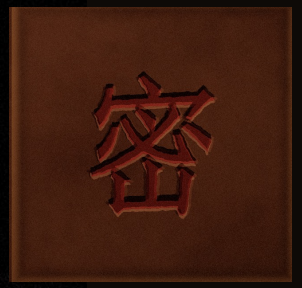
A UV-edge distance–based wear model driven by low-frequency noise simulates oxidation and patina, while bloom, film grain, and vignetting effects unify the visual expression of temporal wear, creating an immersive depiction of the seal's passage through time.

Height Field–Driven Realistic Surface Shading

Engravings via Screen-Space Depth-Field Gradients



Inspect Mode - Seal Inscription Surface Detail



Intaglio Text

Relief Text

Rendering Pipeline · Four-Stage Process

1 Glyph Rasterization into Texture Atla

Each face's Chinese characters are packed into a 6×5 texture atlas, with 28-pixel transparent padding per cell to avoid sampling artifacts.

2 Alpha Channel as Height Map

The fragment shader samples the atlas alpha channel using a 9-point blur kernel to derive local concavity depth.

3 Screen-Space Derivatives for Normal Perturbation

dFdx/dFdy and world-space derivatives generate curvature-aware tangent-space normal perturbations, enabling PBR lighting to bend naturally over surface details.

4 Dual Mode: Intaglio and Relief

A sign toggle controls whether normals are inverted or outward-facing. Concave grooves are enhanced with AO darkening and cinnabar-red accumulation, while edges are subtly brightened to emphasize depth.

From Digital Model to Imprint — The Ritual of Stamping

Stamping Simulation · Press-and-Hold · Mirror-Carved Physics · Timeless Marks

Stamp Mode - UX Design

4-Stage Animation

Descend → Press → Lift → Rest

Ease-in descent, subtle trembling pressure, and ease-out lift, each driven by distinct easing curves to simulate natural variations in human force.

Press-and-Hold Charge Mechanism

Press-and-Hold Charge

Press-and-hold accumulates ink pressure with a charging bar and subtle visual tremor; beyond 80% enters overcharge mode. On release, charge level controls saturation and ink diffusion.

Mirror Engraving — Physically Accurate

Mirror-Carved Transfer

When entering stamping mode, bottom-face characters are automatically mirrored; during transfer onto paper, a second inversion is applied. The two transformations cancel out, producing correctly oriented text on paper while preserving mirrored engraving on the seal body.